Research Methodology

Week 9 Assignment: Questions, Answers and Explanations

Week 9: The Research Journey: Navigating Uncertainty, Setbacks and Discovery

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

The portal records no submission from you for this week, so it shows 0 on every question. The correct answers below are the portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. These questions follow the lecture on the ups and downs of research life.

Answer key at a glance

Question	Type	Correct option(s)	Portal score
1	Select all that apply	**A, B, D**	Not submitted
2	Single answer	**C**	Not submitted
3	Select all that apply	**A, B, C, D**	Not submitted
4	Select all that apply	**A, B, C**	Not submitted
5	Single answer	**C**	Not submitted
6	Select all that apply	**A, B, C**	Not submitted
7	Single answer	**C**	Not submitted
8	Select all that apply	**A, C, D**	Not submitted
9	Single answer	**B**	Not submitted
10	Select all that apply	**A, B, C, D**	Not submitted

Questions with explanations

Question 1 of 10

Which of the following characteristics distinguish a research programme from a regular curriculum-oriented academic programme?

Select all that apply.

Option	Statement	Verdict	Reason
A	**The research question may itself require extensive exploration before becoming well defined.**	**Correct**	In research the problem is often unclear at the start.
B	**Experimental outcomes may deviate from expectations and require modification of the research approach.**	**Correct**	Surprising results can force a change of plan.
C	The programme follows a highly predictable schedule with predefined learning outcomes for every stage.	**Wrong**	This describes a curriculum, not research.
D	**The researcher may repeatedly formulate new questions as insights emerge from previous work.**	**Correct**	Each result raises new questions.

Correct answer: A and B and D.

Explanation: Research is open-ended: the question may need exploration, results can surprise you and new questions keep appearing. A predictable schedule with fixed outcomes at every stage is how a curriculum works.

Question 2 of 10

The lecture compares the research journey to a roller-coaster ride primarily because:

Single answer.

Option	Statement	Verdict	Reason
A	Research requires researchers to work continuously under high pressure.	**Wrong**	The comparison is about ups and downs, not constant pressure.
B	Research consists of predictable stages followed by unavoidable examinations.	**Wrong**	Research stages are not predictable, and exams are not the point.
C	**Research involves alternating periods of progress, uncertainty, setbacks, breakthroughs, and renewed progress.**	**Correct**	A roller coaster goes up and down, and so does research.
D	Research is primarily dependent on collaboration with a large number of researchers.	**Wrong**	Team size is unrelated to the comparison.

Correct answer: C.

Explanation: A roller coaster rises and falls. Research moves in the same way between progress, uncertainty, setbacks, breakthroughs and renewed progress.

Question 3 of 10

According to the lecture, which factors can contribute to the overall research experience of a scholar in a laboratory?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Relationships with advisers and labmates**	**Correct**	Working relationships shape daily life in the lab.
B	**Emotional experiences and friendships**	**Correct**	Feelings and friendships affect motivation.
C	**Availability and accessibility of research facilities**	**Correct**	Equipment and resources shape what can be done.
D	**Support received from friends and family**	**Correct**	Personal support helps a scholar through hard periods.

Correct answer: A and B and C and D.

Explanation: A scholar's experience is shaped by people, emotions, practical facilities and personal support. The lecture lists all four as contributing factors.

Question 4 of 10

Which practices are presented in the lecture as useful for maintaining a sustainable rhythm in research life?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Adequate sleep and physical activity**	Correct	They keep energy and judgment steady.
B	**Taking time for recreation and renewal**	Correct	Rest prevents burnout.
C	**Reflecting periodically on one's activities and decisions**	Correct	Reflection helps correct course.
D	Working continuously for extremely long hours whenever a problem arises	Wrong	This leads to burnout and is the opposite of a sustainable rhythm.

Correct answer: A and B and C.

Explanation: Sleep, exercise, recreation and reflection keep a researcher going over years. Working non-stop whenever a problem appears is not sustainable.

Question 5 of 10

A researcher receives harsh feedback from an adviser and immediately feels angry and personally inadequate. Based on the lecture, what should be the most appropriate immediate response?

Single answer.

Option	Statement	Verdict	Reason
A	Respond immediately to the adviser to defend the research.	Wrong	A reply made in anger is rarely a good one.
B	Suppress the emotion completely and continue working without reflection.	Wrong	Suppressed feelings stay unresolved.
C	**Pause, recognize the emotion, regulate it through techniques such as rhythmic breathing, and then respond with greater clarity.**	Correct	Noticing and calming the emotion first lets you read the feedback calmly and reply well.
D	Discuss the adviser's behaviour with all labmates immediately.	Wrong	It spreads the problem and does not resolve it.

Correct answer: C.

Explanation: The recommended approach is to pause, recognise the emotion and calm it, for example by rhythmic breathing, and only then respond.

Question 6 of 10

Which of the following strategies are specifically discussed in the lecture as ways of processing or reducing intense anger?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Practicing rhythmic breathing**	Correct	Breathing calms the body.
B	**Temporarily stepping away from the situation**	Correct	Distance gives time to cool down.
C	**Journaling what one is feeling**	Correct	Writing the feeling down helps examine it.
D	Immediately confronting the person responsible for the trigger	Wrong	Acting on anger at once usually makes things worse.

Correct answer: A and B and C.

Explanation: Breathing, stepping away and journaling process anger. Immediate confrontation acts on the anger instead of processing it.

Question 7 of 10

A research scholar feels discouraged after a labmate's paper is accepted while their own paper remains unpublished. Which response best reflects the approach advocated in the lecture?

Single answer.

Option	Statement	Verdict	Reason
A	Compare the two researchers and conclude that the labmate is more capable.	Wrong	This judgment helps no one.
B	Suppress the feeling of comparison without examining its cause.	Wrong	Ignoring the feeling leaves it unresolved.
C	**Convert the comparison into curiosity by investigating what the labmate did differently and identifying possible improvements in one's own approach.**	Correct	It turns a negative feeling into something useful.
D	Avoid interacting with the labmate until one's own paper is accepted.	Wrong	It cuts off a source of learning and support.

Correct answer: C.

Explanation: The lecture advocates turning comparison into curiosity: find out what the other person did differently and learn from it.

Question 8 of 10

According to the lecture, prolonged emotional distress or isolation can negatively affect which aspects of a researcher's life?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Research progress**	Correct	Distress slows the work.
B	Only the researcher's publication record	Wrong	The word "only" makes it wrong; the effects are wider.
C	**Physical health**	Correct	Prolonged distress harms the body.
D	**Friendships and social relationships**	Correct	Isolation damages relationships.

Correct answer: A and C and D.

Explanation: Prolonged distress and isolation hurt the research itself, physical health and relationships, so the effects are not limited to the publication record.

Question 9 of 10

A researcher becomes stuck on a difficult problem and begins working for very long hours while withdrawing from friends and labmates. According to the lecture, what is the primary concern with this response?

Single answer.

Option	Statement	Verdict	Reason
A	It demonstrates insufficient commitment to independent research.	Wrong	The behaviour shows too much effort, not too little.
B	**Emotional isolation may worsen the situation and negatively affect research progress as well as personal well-being.**	Correct	Withdrawing removes the support and fresh ideas that help you get unstuck.
C	Long working hours always produce incorrect experimental results.	Wrong	"Always" is an overstatement.
D	Researchers should never attempt to solve difficult problems independently.	Wrong	Trying hard problems alone is not forbidden.

Correct answer: B.

Explanation: The concern is isolation: cutting yourself off from support can make the problem worse and harm both research progress and well-being.

Question 10 of 10

Which of the following constitute potential support systems for a research scholar according to the lecture?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Adviser or supervisor**	**Correct**	An academic guide who can help with the research.
B	**Labmates and friends**	**Correct**	Peers who share the daily experience.
C	**Mentor**	**Correct**	A trusted senior who can give advice.
D	**Family or old friends**	**Correct**	Personal ties outside the lab.

Correct answer: A and B and C and D.

Explanation: Support can come from academic guides (adviser, mentor), peers (labmates, friends) and personal ties (family, old friends). The lecture presents all four as support systems.